Lesson 40: Kinematics with Calculus — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Differentiate $s = 5t^2$.

2. $v = ds/dt$ — what is v?

3. $a = dv/dt$ — what is a?

4. 'At rest' means which quantity is 0?

5. $s = 4t^2$. Find v.

6. $v = 6t - 2$. Find a.

Section 2 — Core practice

7. $s = t^3 - 6t^2$. Find v.

8. $s = t^3 - 6t^2$. Find a.

9. $v = 4t - 8$. When is the object at rest?

10. $s = 2t^3$. Find v and a.

11. $v = 3t^2 - 3$. When at rest ($t \geq 0$)?

12. $s = t^2 - 10t$. Find v and when $v = 0$.

Section 3 — Challenge & exam-style

13. $s = 2t^3 - 9t^2 + 12t$. Find v and the times when the particle is at rest.

14. $s = t^3 - 3t^2 - 9t$. Find the times at rest ($t \geq 0$).

15. $v = t^2 - 5t + 6$. Find the acceleration when $t = 4$.

16. $s = t^3 - 12t$. Find the velocity and acceleration at $t = 2$.